SK52~SK510(SS52~SS510) SMC

Schottky rectifier

Major Ratings and Characteristics

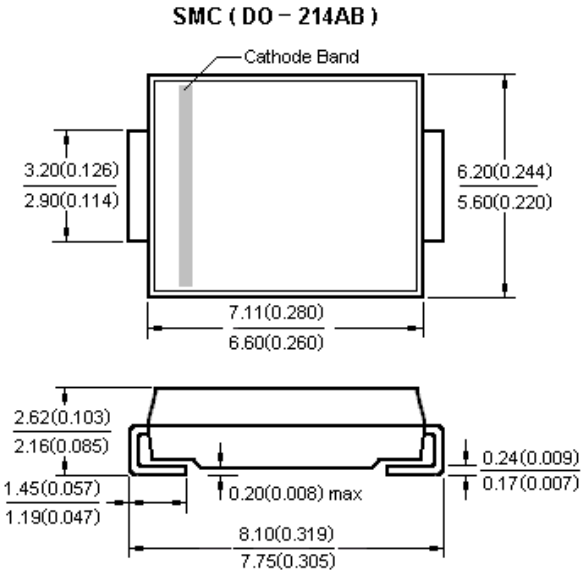
$I_{F(AV)}$	5.0 A
V_{RRM}	20 V to 100 V
I_{FSM}	150 A
V_F	0.55 V , 0.70 V, 0.85V
T_j max.	150 °C



SMC (DO – 214AB)

Features

- Low profile package
- Ideal for automated placement
- Ultrafast reverse recovery time
- Low power losses, high efficiency
- Low forward voltage drop
- High surge capability
- High temperatrue soldering：
  260℃/10 seconds at terminals
- Component in accordance to
  RoHS 2002/95/1 and WEEE 2002/96/EC



Dimentsions in millimeters and (inchs)

Mechanical Date

- Case: JEDEC DO-214ABmolded plastic body over passivated chip
- Terminals: Solder plated, solderable per J-STD-002B and JESD22-B102D
- Polarity: Laser band denotes cathode end

Maximum Ratings & Thermal Characteristics & Electrical Characteristics

(TA = 25 °C unless otherwise noted)

	Symbol	SK52 SS52	SK53 SS53	SK54 SS54	SK55 SS55	SK56 SS56	SK58 SS58	SK510 SS510	UNIT
Maximum repetitive peak reverse voltage	V_{RRM}	20	30	40	50	60	80	100	V
Maximum RMS voltage	V_{RMS}	14	21	28	35	42	56	70	V
Maximum DC blocking voltage	V_{DC}	20	30	40	50	60	80	100	V
Maximum average forward rectified current	$I_{F(AV)}$	5							A
Peak forward surge current 8.3 ms single half sine-wave superimposed on rated load	I_{FSM}	150							A
Maximum instantaneous forwad voltage at 5.0A	V_F	0.55			0.70		0.85		V
Maximum DC reverse current T_A = 25 ℃	I_R	0.5							mA
at Rated DC blocking voltage T_A = 100℃		10					20		mA
Voltage rate of change (rated VR)	dv/dt	10000							V/μs
Thermal resistance from junction to ambient	$R_{\theta JA}$	88							℃/ W
Operating junction and storage temperature range	T_J, T_{STG}	–65 to +150							℃

SK52~SK510(SS52~SS510) SMC

Schottky rectifier

Characteristic Curves (T_A=25 ℃ unless otherwise noted)

Fig.1 Forward Current Derating Curve

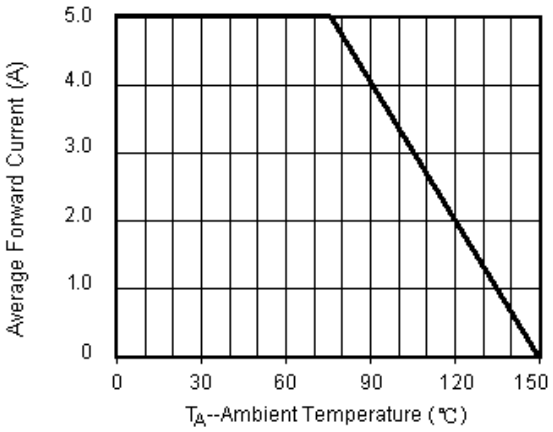


Fig.2 Maximum Non-Repetitive Peak Forward Surge Current

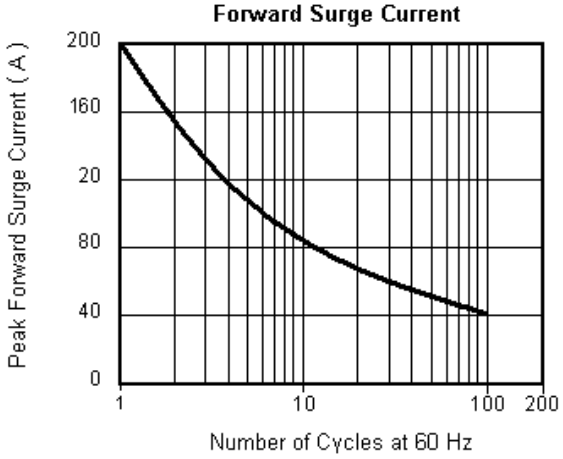


Fig.3 Typical Instantaneous Forward Characteristics

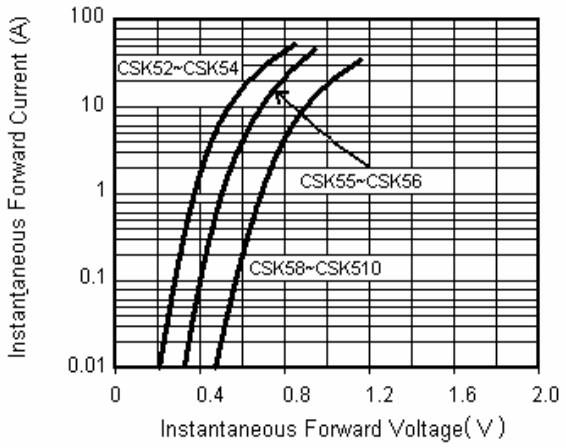


Fig.4 Typical Reverse Leakage Characteristics

